

Unit 2

ATOMS, MOLECULES AND STOICHIOMETRY

39 items · 86 marks

Syllabus sections in this unit

- 2.1 Relative masses of atoms and molecules
- 2.2 The mole and the Avogadro constant
- 2.3 Formulas
- 2.4 Reacting masses and volumes (of solutions and gases)

Unit 2: Atoms, molecules and stoichiometry

1. M/J 26/P21/Q1(e)

(e) Compound **X** contains sulfur and chlorine only.

100.0 g of **X** contains 47.49 g of sulfur. The relative molecular mass, M_r , of **X** is 135.2.

Determine the molecular formula of **X**. Show your working.

molecular formula of **X** = [3]

2. M/J 26/P22/Q4(b)(ii)-(iii)

Context: part (b) (introduction)

(b) The structure of compound **X** is shown in Figure 4.2.

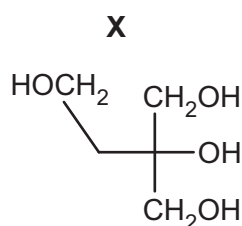


Figure 4.2

The molecular formula of **X** is $C_5H_{12}O_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of 0.500 mol dm^{-3} acidified $K_2Cr_2O_7(aq)$.

(ii) Calculate the amount, in mol, of $K_2Cr_2O_7(aq)$ used in the reaction.

amount = mol [1]

(iii) Calculate the amount, in mol, of **X** present in the 0.255 g sample.

amount = mol [1]

3. M/J 26/P24/Q1(b)

Context

1 Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

(b) Deduce the formula of aluminium nitrate(V).

[1]

4. O/N 25/P21/Q3(a)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

[1]

5. O/N 25/P23/Q3(a)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

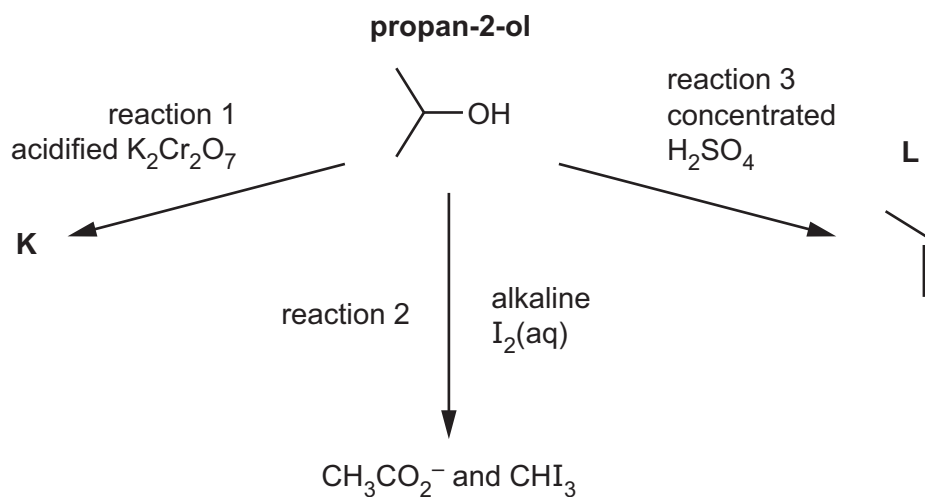
[1]

6. O/N 25/P24/Q3(b)(iv)

Context

- 3** Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

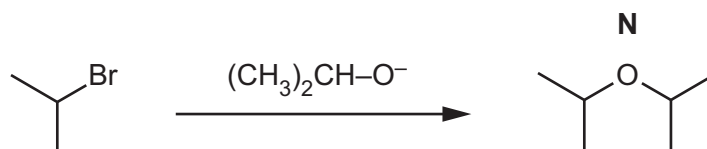
Fig. 3.1 shows some reactions of propan-2-ol.

**Fig. 3.1**

Context: part (b) (introduction)

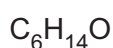
- (b) Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH}-\text{O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

**Fig. 3.3**

- (iv) **N** is also added to petrol to make it burn more smoothly.

Construct an equation for the complete combustion of **N**, $\text{C}_6\text{H}_{14}\text{O}$.



[1]

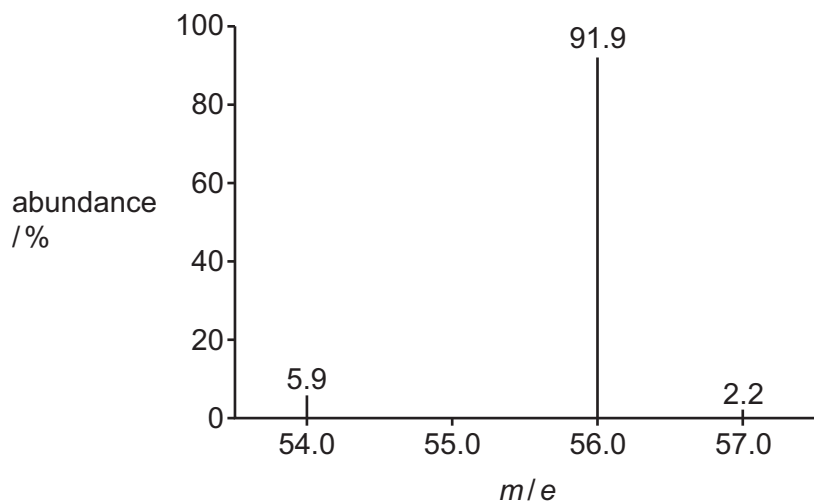
7. M/J 25/P21/Q2(b)(i)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

Context: part (b) (introduction)

- (b)** Fig. 2.1 shows the mass spectrum of the sample of iron.

**Fig. 2.1**

- (i)** Use Fig. 2.1 to calculate the relative atomic mass, A_r , of iron to **one** decimal place.

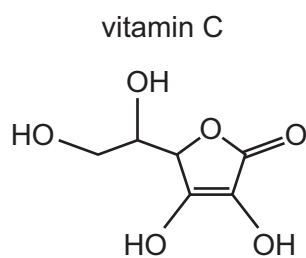
Show your working.

A_r of iron = [2]

8. M/J 25/P21/Q4(a)

Context

- 4** The structure of vitamin C is shown in Fig. 4.1.

**Fig. 4.1**

- (a)** Deduce the empirical formula of vitamin C.

[1]

9. M/J 25/P21/Q4(b)(i)-(ii)

Context

4 The structure of vitamin C is shown in Fig. 4.1.

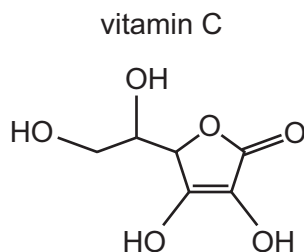


Fig. 4.1

Context: part (b) (introduction)

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

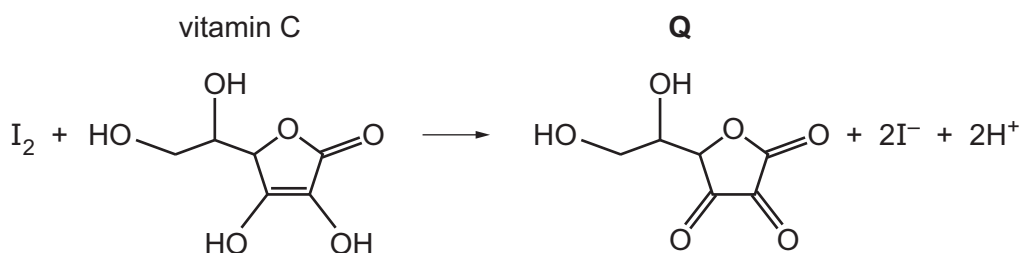


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm^3 of vitamin C solution. 5.00 cm^3 of this vitamin C solution is added to a flask with approximately 150 cm^3 of water and an indicator. Exactly 28.40 cm^3 of $5.00 \times 10^{-4}\text{ mol dm}^{-3}\text{ I}_2(\text{aq})$ reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(i) Calculate the amount, in mol, of $\text{I}_2(\text{aq})$ added to the flask in this titration.

amount of $\text{I}_2(\text{aq}) =$ mol [1]

(ii) Use your answer to **(b)(i)** to calculate the mass, in g, of vitamin C in the tablet. Show your working.

(If you were unable to calculate a value for the amount of $\text{I}_2(\text{aq})$ in **(b)(i)**, use the value $2.64 \times 10^{-4} \text{ mol}$. This is **not** the correct value.)

mass of vitamin C in tablet = g [2]

10. M/J 25/P22/Q2(a)

Context

2 A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(a) (i) Define relative atomic mass.

[2]

(ii) Table 2.1 shows the abundances of two of the isotopes present in the sample of iron.

Table 2.1

isotope	relative isotopic mass	abundance / %
^{54}Fe	53.9	6.0
^{56}Fe	55.9	91.9

Use Table 2.1 to calculate the relative isotopic mass of the third isotope of iron in the sample. Show your working.

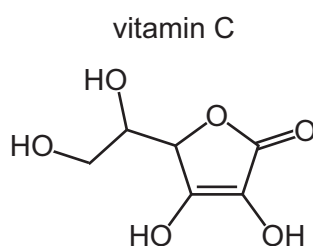
relative isotopic mass =

[2]

11. M/J 25/P22/Q5(a)

Context

5 The structure of vitamin C is shown in Fig. 5.1.

**Fig. 5.1**

(a) Deduce the empirical formula of vitamin C.

[1]

Context

vitamin C

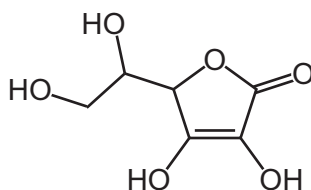


Fig. 5.1

Context: part (b) (introduction)

stage 1 All the vitamin C in 150.0g of lemon is extracted and dissolved in water to make 100.0 cm³ of solution L.

stage 2 A 25.0 cm³ sample of solution **L** is titrated with 5.00 × 10⁻³ mol dm⁻³ I₂(aq).

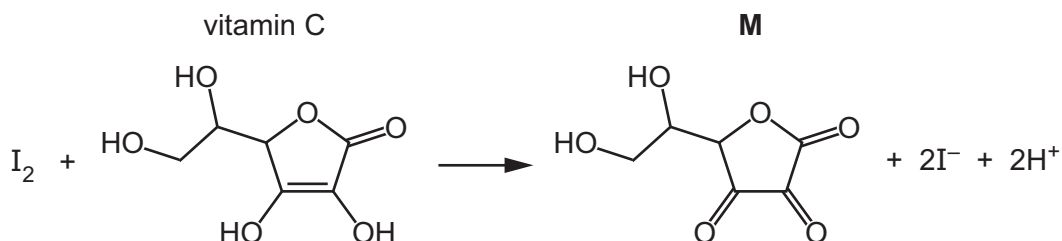


Fig. 5.2

Exactly 36.65 cm^3 of $\text{I}_2(\text{aq})$ reacts with the 25.0 cm^3 sample of solution L.

[M_r : vitamin C, 176]

(i) Calculate the amount, in mol, of $\text{I}_2(\text{aq})$ added in the titration.

amount of $\text{I}_2(\text{aq}) =$ mol [1]

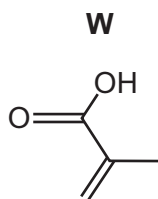
(ii) Use your answer to (b)(i) to calculate the percentage by mass of vitamin C present in 150.0 g of lemon.

(If you were unable to calculate a value for the amount of $\text{I}_2(\text{aq})$ in **(b)(i)**, use the value $7.65 \times 10^{-4} \text{ mol}$. This is **not** the correct value.)

percentage by mass = % [2]

13. M/J 25/P23/Q5(a)(i)

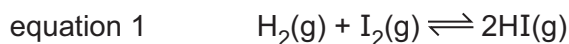
Context: part (a) (introduction)

5 (a) W is a colourless liquid.**Fig. 5.1****(i)** Deduce the empirical formula of **W**.

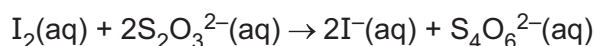
[1]

14. M/J 25/P24/Q3(a)

Context

3 $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.

In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

(a) In experiment 1, $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ are combined.Water is then added to the equilibrium mixture to produce 1.00 dm^3 of solution **A**.The amount of $\text{I}_2(\text{aq})$ present in 1.00 dm^3 of solution **A** is found by titration with $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$.Exactly 32.90 cm^3 of $0.200 \text{ mol dm}^{-3}$ $\text{Na}_2\text{S}_2\text{O}_3$ reacts with all of the I_2 in a 25.0 cm^3 sample of solution **A**.**(i)** Calculate the amount, in mol, of $\text{S}_2\text{O}_3^{2-}$ that reacts in the titration.

amount of $\text{S}_2\text{O}_3^{2-} =$ mol [1]

(ii) Use your answer to **(a)(i)** to calculate the amount, in mol, of I_2 present in 1.00 dm^3 of solution **A**.

(If you were unable to obtain an answer in **(a)(i)**, then use $9.42 \times 10^{-3} \text{ mol}$. This is **not** the correct answer.)

amount of I_2 in 1.00 dm^3 of solution **A** = mol [2]

15. M/J 25/P24/Q6(a)(i)-(ii)

Context

6 **P**, **Q** and **R** are three different hydrocarbon molecules.**(a) (i)** Define empirical formula.

[1]

(ii) Hydrocarbon **P** contains 85.7% by mass of carbon.Calculate the empirical formula of **P**.

Show your working.

empirical formula of **P** =

[2]

16. O/N 24/P21/Q1(c)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.**(c)** Table 1.2 shows the relative abundances of isotopes in a sample of an alloy containing rhodium and iridium only.**Table 1.2**

isotope	relative isotopic mass	relative abundance in alloy
$^{103}_{45}\text{Rh}$	102.91	50.00
$^{191}_{77}\text{Ir}$	190.96	15.18
$^{193}_{77}\text{Ir}$	192.96	34.82

(i) Define relative isotopic mass.

[2]

(ii) Use Table 1.2 to calculate the relative atomic mass, A_r , of iridium in the alloy.Give your answer to **two** decimal places.

relative atomic mass of iridium =

[2]

17. O/N 24/P21/Q1(d)(i)

Context

- 1** Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

Context: part (d) (introduction)

- (d)** Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

- (i)** Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$.

Show your working.

$$x =$$

[2]

18. O/N 24/P22/Q1(b)(ii)-(iii)

Context

- 1** Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

Context: Table 1.1

Table 1.1

isotope	relative isotopic mass	number of protons	number of neutrons
${}^{93}_{41}\text{Nb}$	92.91		
${}^{181}_{73}\text{Ta}$	180.95		

Context: part (b) (introduction)

- (b)** Pelopium was the suggested name for a new element discovered in a mineral.

Pelopium was later found to be a mixture of niobium, Nb, and tantalum, Ta.

Only one naturally occurring isotope exists for each of Nb and Ta.

- (ii)** Define relative isotopic mass.

[2]

- (iii)** A sample of pelopium contains 90.9% by mass ${}^{93}_{41}\text{Nb}$ and 9.1% by mass ${}^{181}_{73}\text{Ta}$.

Calculate the theoretical relative atomic mass of pelopium based on these data and Table 1.1.

Give your answer to two decimal places.

Show your working.

theoretical relative atomic mass of pelopium =

[2]

19. O/N 24/P23/Q1(c)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(c) Table 1.2 shows the relative abundances of isotopes in a sample of an alloy containing rhodium and iridium only.

Table 1.2

isotope	relative isotopic mass	relative abundance in alloy
$^{103}_{45}\text{Rh}$	102.91	50.00
$^{191}_{77}\text{Ir}$	190.96	15.18
$^{193}_{77}\text{Ir}$	192.96	34.82

(i) Define relative isotopic mass.

[2]

(ii) Use Table 1.2 to calculate the relative atomic mass, A_r , of iridium in the alloy.

Give your answer to **two** decimal places.

relative atomic mass of iridium =

[2]

20. O/N 24/P23/Q1(d)(i)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

Context: part (d) (introduction)

(d) Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$.

Show your working.

$x =$

[2]

21. M/J 24/P21/Q2(d)(i)

Context: part (d) (introduction)

(d) Boron nitride is a white solid that melts above 2900 °C.

Fig. 2.2 shows part of the lattice structure of a crystal of boron nitride.

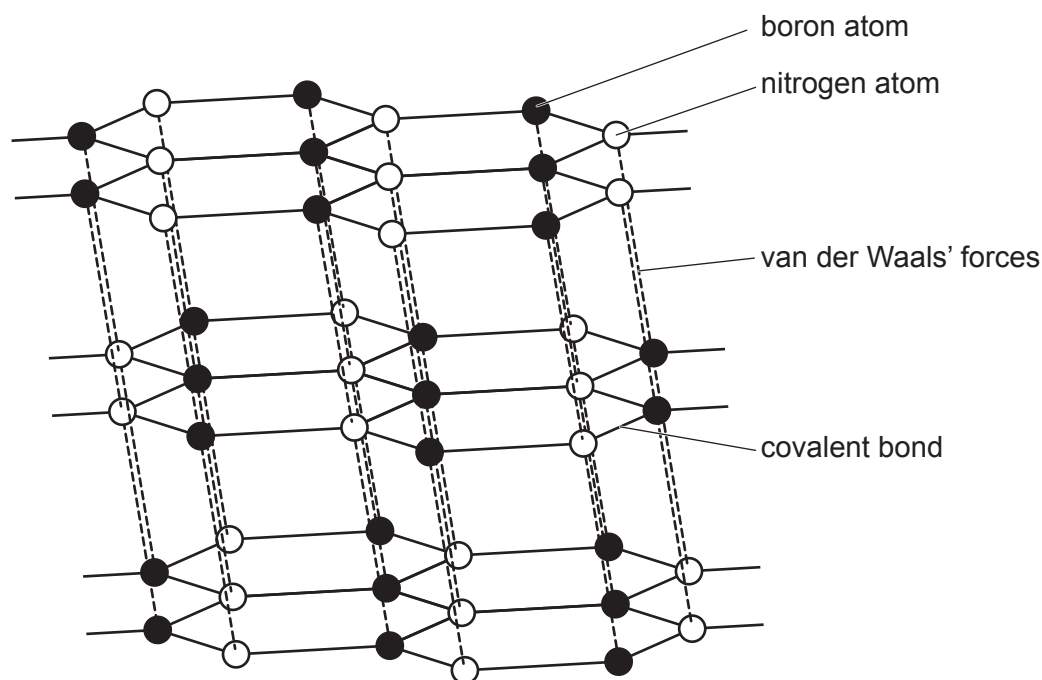


Fig. 2.2

(i) Use Fig. 2.2 to deduce the empirical formula of boron nitride.

[1]

22. M/J 24/P23/Q3(c)(ii)

Context: part (c) (introduction)

(c) Boron carbide is a hard crystalline solid that has a melting point greater than 2000 °C.

(ii) 100 g of pure boron carbide contains 78.26 g of boron.

Calculate the empirical formula of boron carbide.

Show your working.

empirical formula of boron carbide

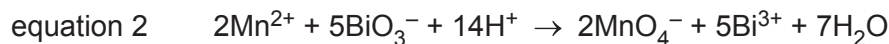
[2]

23. MAR 24/P22/Q1(e)

Context

1 Bismuth is an element in Group 15 of the Periodic Table.

- (e) NaBiO_3 can be used to determine the concentration of $\text{Mn}^{2+}(\text{aq})$. The ionic equation for the reaction is shown in equation 2.



A student uses the following procedure in an experiment.

- Add 100.0 cm^3 of a saturated solution of $\text{Mn}^{2+}(\text{aq})$ to a volumetric flask.
- Add distilled water to the flask to make a 1.00 dm^3 diluted solution.
- Titrate a 25.00 cm^3 sample of the diluted solution with 0.100 mol dm^{-3} $\text{NaBiO}_3(\text{aq})$.

The 25.00 cm^3 sample of the diluted solution of $\text{Mn}^{2+}(\text{aq})$ reacts completely with exactly 21.50 cm^3 of 0.100 mol dm^{-3} $\text{NaBiO}_3(\text{aq})$.

Calculate the concentration, in mol dm^{-3} , of $\text{Mn}^{2+}(\text{aq})$ in the saturated solution.

Show your working.

concentration of $\text{Mn}^{2+}(\text{aq})$ in the saturated solution = mol dm^{-3} [3]

24. M/J 23/P21/Q2(b)(i)

Context

2 A neutralisation reaction occurs when $\text{NaOH}(\text{aq})$ is added to $\text{H}_2\text{SO}_4(\text{aq})$.



Context: part (b) (introduction)

- (b) An experiment is carried out to calculate ΔH_{neut} for the reaction between $\text{NaOH}(\text{aq})$ and $\text{H}_2\text{SO}_4(\text{aq})$.

100 cm^3 of 1.00 mol dm^{-3} $\text{NaOH}(\text{aq})$ is added to 75 cm^3 of 1.00 mol dm^{-3} $\text{H}_2\text{SO}_4(\text{aq})$ in a polystyrene cup and stirred. Results from the experiment are shown in Table 2.1.

Table 2.1

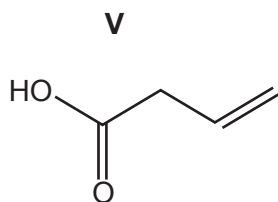
initial temperature of $\text{NaOH}(\text{aq})/^\circ\text{C}$	20.0
initial temperature of $\text{H}_2\text{SO}_4(\text{aq})/^\circ\text{C}$	20.0
maximum temperature of mixture/ $^\circ\text{C}$	27.8

- (i) Use equation 1 to calculate the amount, in mol, of $\text{H}_2\text{SO}_4(\text{aq})$ that is neutralised in the experiment.

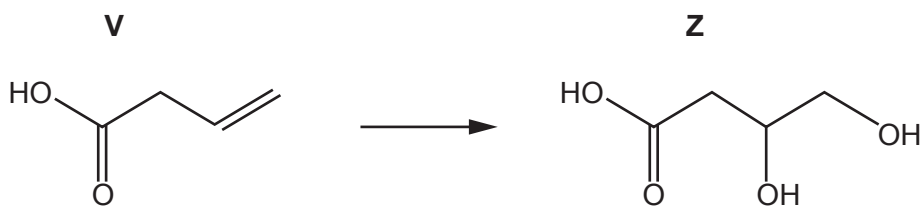
amount of $\text{H}_2\text{SO}_4(\text{aq})$ neutralised = mol [1]

25. M/J 23/P21/Q4(b)(ii)

Context

4 **V** is a colourless liquid.**Fig. 4.1**

Context: part (b) (introduction)

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.**Fig. 4.2****(ii)** Deduce the empirical formula of **Z**.

[1]

26. M/J 23/P22/Q2(e)(iii)

Context

2 Chlorine is a reactive element. It forms many compounds.

Context: part (e) (introduction)

(e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaClO , to water.**(iii)** A sample of bleach **W** contains an unknown concentration of sodium chlorate(I).

10.0 cm^3 of **W** is diluted with distilled water to make a total volume of 100 cm^3 of bleach solution. 25.0 cm^3 of this diluted bleach solution is added to an excess of hydrogen peroxide and the volume of gas produced measured under room conditions. The experiment is repeated and on average 25.0 cm^3 of diluted bleach solution produces 42.0 cm^3 of gas.

Calculate the concentration, in g dm^{-3} , of sodium chlorate(I) in **W**.

concentration of NaClO in **W** = g dm^{-3} [3]

27. M/J 23/P23/Q1(b)

Context

- 1** Copper is used in electrical equipment. It has a melting point of 1085 °C.
- (b)** The relative isotopic masses and natural abundances of the two isotopes in a sample of copper are shown in Table 1.1.

Table 1.1

isotope	relative isotopic mass	% abundance
^{63}Cu	62.930	69.15
^{65}Cu	64.928	30.85

- (i)** Define the unified atomic mass unit. [1]
- (ii)** Define relative atomic mass, A_r , in terms of the unified atomic mass unit. [1]
- (iii)** Calculate the relative atomic mass, A_r , of copper in this sample using the data in Table 1.1.

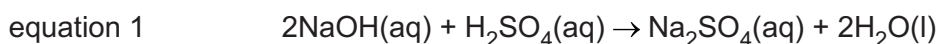
Show your working.

$$A_r = \quad [1]$$

28. M/J 23/P23/Q3(b)(i)

Context

- 3** A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



Context: part (b) (introduction)

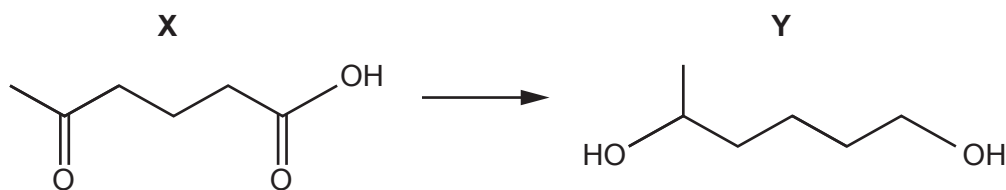
- (b)** In an experiment, 50.0 cm³ of 2.00 mol dm⁻³ NaOH(aq) is added to 60.0 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Both solutions have a temperature of 21.4 °C before mixing. The maximum temperature of the mixture is measured.
- (i)** Use equation 1 to calculate the amount, in mol, of H₂SO₄(aq) that is neutralised in the experiment.

$$\text{amount of H}_2\text{SO}_4 \text{ neutralised} = \quad \text{mol} \quad [1]$$

29. M/J 23/P23/Q5(a)

Context

- 5 Y is formed from X in a single-step reaction, as shown in Fig. 5.1.

**Fig. 5.1**

- (a) Deduce the empirical formula of Y.

[1]

30. MAR 23/P22/Q2(d)(ii)

Context

- 2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

- (ii) Compound X, an anhydrous Group 2 bromide, is dissolved in water and titrated against aqueous silver nitrate.

A solution containing 0.250 g of X requires 33.65 cm³ of 0.0500 mol dm⁻³ AgNO₃(aq) for complete reaction.

Identify X.

Show your working.

X =

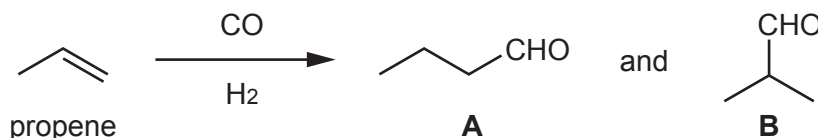
[3]

31. MAR 23/P22/Q3(a)(iv)

Context

- 3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

- (iv) The complete reaction of propene with a 1:1 mixture of CO and H₂ produces A and B only. The product mixture contains 96% A and 4% B.

Calculate the mass of A produced in this reaction when 5.00 × 10³ kg of propene is used.

mass of A =

kg [1]

32. O/N 22/P21/Q1(a)(ii)-(iii)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

Context: part (a) (introduction)

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

(ii) Define relative atomic mass.

[2]

(iii) The relative atomic mass of gallium, A_r , is 69.723.
The relative isotopic masses of ^{69}Ga and ^{71}Ga are:

^{69}Ga , 68.926; ^{71}Ga , 70.925.

Use this information to calculate the percentage abundance of ^{69}Ga in elemental gallium.
Show your working.

Assume that the element contains only the ^{69}Ga and ^{71}Ga isotopes.
Give your answer to **four** significant figures.

percentage abundance of ^{69}Ga = %
[2]

33. O/N 22/P22/Q1(d)

Context

- 1** Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(d) There are many naturally occurring hydrated compounds that contain the anion PO_4^{3-} .

(i) Name the anion PO_4^{3-} .

[1]

(ii) Struvite is a soft hydrated mineral with $M_r = 245.3$. The anhydrous form of the mineral has the formula NH_4MgPO_4 .

Calculate the number of molecules of water of crystallisation in struvite.

Give your answer to the nearest integer. Show your working.

number of molecules of water of crystallisation = [2]

34. O/N 22/P23/Q1(a)(ii)-(iii)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

Context: part (a) (introduction)

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

(ii) Define relative atomic mass.

[2]

(iii) The relative atomic mass of gallium, A_r , is 69.723.
The relative isotopic masses of ^{69}Ga and ^{71}Ga are:

^{69}Ga , 68.926; ^{71}Ga , 70.925.

Use this information to calculate the percentage abundance of ^{69}Ga in elemental gallium.
Show your working.

Assume that the element contains only the ^{69}Ga and ^{71}Ga isotopes.
Give your answer to **four** significant figures.

percentage abundance of ^{69}Ga = %
[2]

35. M/J 22/P21/Q1(e)(i)-(ii)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

Context: part (e) (introduction)

(e) A sample of magnesium contains three isotopes, ^{25}Mg , ^{26}Mg and **X**.

The percentage abundance of the three isotopes is shown in Table 1.1.

Table 1.1

isotope of Mg	mass/a.m.u.	percentage abundance/%
X		78.99
^{25}Mg	24.99	10.00
^{26}Mg	25.98	11.01

- (i)** The relative atomic mass, A_r , is calculated by comparing the average mass of the isotopes of an element to the unified atomic mass unit.

Define the unified atomic mass unit.

[1]

- (ii)** Calculate the mass of **X**. Use data from Table 1.1 and A_r (magnesium) = 24.31 in your calculation. Show your working.

mass of **X** =

[2]

36. M/J 22/P21/Q4(a)

Context

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

(a) Calculate the molecular formula of **V**. Show your working.

molecular formula of **V** = [3]

37. M/J 22/P21/Q4(b)(ii)

Context

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

Context: part (b) (introduction)

(b) **V** contains two types of functional group: a carboxylic acid and an alkene.

(ii) A 3.196 g sample of Br_2 reacts completely with 2.800 g of **V**.

Calculate how many alkene functional groups are present in one molecule of **V**. Show your working.

number of alkene functional groups in **V** = [1]

38. M/J 22/P22/Q6(b)

Context

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

(b) Table 6.2 shows the percentage by mass of each element present in **Z**.

Table 6.2

element	percentage by mass/%
carbon	41.38
hydrogen	3.45
oxygen	55.17

Using the data in Table 6.2, demonstrate that the empirical formula of **Z** is CHO . Show your working.

[1]

39. M/J 22/P23/Q1(f)(i)

(f) (i) The element copper has a relative atomic mass of 63.5.

Calculate how many atoms are present in 1.05 g of copper.

atoms of copper present = [1]

Answers Section

1. M/J 26/P21/Q1(e)

1(e)	M1 mass $Cl = 52.51$ M2 $47.49 / 32.1 : 52.51 / 35.5 (= 1.48 : 1.48)$ (empirical formula =) SCl M3 ($M_r(X) = 135.2$) molecular formula = S_2Cl_2	3 check working M1 allow $100 - 47.49$ without being evaluated A owtte $135.2 - \times(32.1) s = 2$ so $M_r = S_2Cl_2$ ✓✓✓ A alternative methods M2 A ecf from an incorrect M1 M3 A only S_2Cl_2 (with no working)
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2. M/J 26/P22/Q4(b)(ii)-(iii)

4(b)(ii)	0.005 OR 5×10^{-3}	1	I 1 / 200
4(b)(iii)	1.875×10^{-3}	1	A 1.88×10^{-3} OR 1.9×10^{-3} A 2×10^{-3} with appropriate working R 1.880×10^{-3} OR 1.90×10^{-3} OR 2.0×10^{-3} OR 1.87×10^{-3}

3. M/J 26/P24/Q1(b)

1(b)	$Al(NO_3)_3$	1	A $(NO_3)_3Al$
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4. O/N 25/P21/Q3(a)

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	1
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5. O/N 25/P23/Q3(a)

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	1
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6. O/N 25/P24/Q3(b)(iv)

3(b)(iv)	$C_6H_{14}O + 9O_2 \rightarrow 6CO_2 + 7H_2O$	1
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7. M/J 25/P21/Q2(b)(i)

2(b)(i)	M1 method showing each abundance multiplied by m/e value (5.9×54) and (91.9×56) and (2.2×57) OR (0.59×54) and (9.19×56) and (0.22×57) M2 the three values from M1 expressed as a percentage and added together correctly with final answer to 1 d.p $\frac{5.9}{100} \times 54 + \frac{91.9}{100} \times 56 + \frac{2.2}{100} \times 57 = 55.9$	2
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8. M/J 25/P21/Q4(a)

4(a)	$C_3H_4O_3$	1
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9. M/J 25/P21/Q4(b)(i)-(ii)

4(b)(i)	$(28.40 / 1000 \times 0.0005) = 1.42 \times 10^{-5} \text{ mol}$	1
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4(b)(ii)	$(1.42 \times 10^{-5} \times 176 \times 200 / 5) = 0.099968$ M1 correct method to deduce mass of vit C in 5 cm ³ sample $(1.42 \times 10^{-5} \times 176) = 2.49 \times 10^{-3}$ M2 correct calculation for mass of vit C in original solution $(2.49 \times 10^{-3} \times 200 / 5) = 0.099968$ OR M1 correct method to deduce amount, mol, vit C in 5 cm ³ sample $(1.42 \times 10^{-5} \times 200 / 5) = 5.68 \times 10^{-4}$ M2 correct calculation for mass of vit C in original solution = M1 $\times$ 176 $(5.68 \times 10^{-4}) \times 176 = 0.099968$	2
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10. M/J 25/P22/Q2(a)

2(a)(i)	route 1 M1 average (weighted) mass of (an) atom(s) (of an element) OR average (weighted) mass of the isotope(s) (of an element) M2 compared to the mass of the unified atomic mass unit OR on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 (units) OR on a scale in which the mass of a carbon-12 atom/isotope is exactly 12 (units) OR compared to $\frac{1}{12}$ mass of one atom of C-12 route 2 M1 average mass of one mol of isotopes / atoms (of an element) M2 relative / compared to $\frac{1}{12}$ of the mass of one mol of C-12 atoms/isotopes OR when the mass of one mol C-12 atoms / isotopes is exactly 12 (000) g	2
2(a)(ii)	M1 relative abundance of unknown isotope = $(100 - 91.9 - 6.0) = 2.1\%$ M2 $55.8 = (53.9 \times 6.0 + 55.9 \times 91.9 + ? \times M1) / 100 = 56.9$	2

11. M/J 25/P22/Q5(a)

5(a)	C ₃ H ₄ O ₃	1
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12. M/J 25/P22/Q5(b)(i)-(ii)

5(b)(i)	$5.00 \times 10^{-3} \times 36.65 / 1000$ $= 1.833 \times 10^{-4} / 1.8325 \times 10^{-4}$	1
5(b)(ii)	M1 expression and / or correct calculation to show how the amount of vitamin C in 150 g of lemon peel is calculated $1.8325 \times 10^{-4} \text{ mol} \times 176 \times 4$ M2 correct calculation based on $(M1 / 150) \times 100 = 0.086(\%)$ Using 1.8325×10^{-4} for M1 , M2 = 0.086 OR 0.087(%) Alternative answer using 7.65×10^{-4} : M1 $7.65 \times 10^{-4} \times 176 \times 4 = 0.5386 / 0.539$ M2 correct calculation based on $(M1 / 150) \times 100 = 0.359 / 0.36 (\%)$	2

13. M/J 25/P23/Q5(a)(i)

5(a)(i)	C ₂ H ₃ O	1
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14. M/J 25/P24/Q3(a)

3(a)(i)	0.00658(0) mol	1
3(a)(ii)	M1 I ₂ in 25.0 cm ³ $0.00658 / 2 = 0.00329$ (mol) M2 $0.00329 \times 1000 / 25.0 = 0.132$ (mol) alternative answer using 9.42×10^{-3} : M1 0.00471 M2 0.1884	2

15. M/J 25/P24/Q6(a)(i)-(ii)

6(a)(i)	simplest (whole number) ratio of atoms (of the elements) in a compound	1
6(a)(ii)	M1 M2 $\begin{array}{ccc} \text{C} & : & \text{H} \\ \frac{85.7}{12} \text{ or } 7.14 & : & \frac{14.3}{1} \text{ or } 14.3 \\ 1 & : & 2 \end{array}$ CH_2 2	2

16. O/N 24/P21/Q1(c)

1(c)	mass of an atom of an isotope	1
	compared to the mass of the unified atomic mass unit (amu) OR on a scale on which a carbon-12 atom has a mass of exactly 12 units OR divided by $\frac{1}{12}$ mass of a carbon-12 atom	1
1(c)(ii)	$\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$	1
	= 192.35	1

17. O/N 24/P21/Q1(d)(i)

1(d)(i)	M1 $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$	1
	M2 $x = 3$	1

18. O/N 24/P22/Q1(b)(ii)-(iii)

1(b)(ii)	M1 mass of an (atom of an) isotope	1
	M2 relative/compared to (the mass of) the unified atomic mass unit OR on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 units OR divided by / compared to $\frac{1}{12}$ of the mass of a carbon-12 atom / isotope Alternative route using mass of 1 mol throughout M1 mass of one mol of an (atom of an) isotope M2 relative/compared to $\frac{1}{12}$ of the mass of 1 mol of C-12 atom / isotope OR when the mass of one mol C-12 atom / isotope is exactly 12(.000) g	1
1(b)(iii)	$92.91 \times 0.909 + 180.95 \times 0.091$	1
	= 100.92	1

19. O/N 24/P23/Q1(c)

1(c)	mass of an atom of an isotope	1
	compared to the mass of the unified atomic mass unit (amu) OR on a scale on which a carbon-12 atom has a mass of exactly 12 units OR divided by $\frac{1}{12}$ mass of a carbon-12 atom	1
1(c)(ii)	$\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$	1
	= 192.35	1

20. O/N 24/P23/Q1(d)(i)

1(d)(i)	M1 $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$	1
	M2 $x = 3$	1

21. M/J 24/P21/Q2(d)(i)

2(d)(i)	BN	1
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22. M/J 24/P23/Q3(c)(ii)

3(c)(ii)	M1 give correct expression or calculate mol B compared to mol C $78.26 / 10.8 (= 7.246 / 7.25 \text{ mol B})$ AND $21.74 / 12 (= 1.81(2) \text{ mol C})$ M2 Find B : C ratio AND use this to state empirical formula $7.246 \text{ to } 7.25 / 1.817 \text{ to } 1.82 = 3.99 \text{ to } 3.98 \text{ mol B : 1 mol C}$ (so) B ₄ C OR CB ₄	2
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23. MAR 24/P22/Q1(e)

1(e)	moles of NaBiO ₃ = $21.50 / 1000 \times 0.100$ (= $2.15 \times 10^{-3} \text{ mol}$)	1
	moles of Mn ²⁺ in dil. = $2.15 \times 10^{-3} \times 2 / 5 \times 40$ (= 0.0344 mol)	1
	conc. saturated Mn ²⁺ = $0.0344 \times 1000 / 100$ = 0.344 mol dm⁻³	1

24. M/J 23/P21/Q2(b)(i)

2(b)(i)	0.05 mol	1
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25. M/J 23/P21/Q4(b)(ii)

4(b)(ii)	CH ₂ O	1
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26. M/J 23/P22/Q2(e)(iii)

2(e)(iii)	M1 (show how to calculate) mol O ₂ gas 42 cm ³ gas = 42 / 24000 mol oxygen gas (= 1.75 × 10 ⁻³ mol) M2 (process to find) amount NaClO in 10 cm ³ sample = 100 cm ³ (diluted) bleach M1 × 4 (= 4 × 1.75 × 10 ⁻³ mol in 10 cm ³ = 0.007) M3 correct calculation using M2 to find concentration = (M2 / 10) × 1000 × 74.5 = 52.2 (g / dm ³)	3
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27. M/J 23/P23/Q1(b)

1(b)(i)	(the mass of) one twelfth of the mass of a C-12 / ¹² C atom (or isotope)	1
1(b)(ii)	average mass of the isotopes of an element compared to the unified atomic mass unit	1
1(b)(iii)	(62.93 × 69.15) ÷ 100 + (64.928 × 30.85) ÷ 100 = 63.55 / 63.546	1

28. M/J 23/P23/Q3(b)(i)

3(b)(i)	0.05 mol (H ₂ SO ₄ (aq))	1
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29. M/J 23/P23/Q5(a)

5(a)	C ₃ H ₇ O	1
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30. MAR 23/P22/Q2(d)(ii)

2(d)(ii)	moles of AgNO ₃ = $\frac{33.65}{1000} \times 0.0500$ (= 1.68(25) × 10 ⁻³)	1
	moles of X = ½ × moles of AgNO ₃ = 0.250 ÷ M _r (X) ∴ M _r (X) = 297.2	1
	A _r of Group 2 element is 297.2 – 2(79.9) = 137.4 AND X is BaBr ₂	1

31. MAR 23/P22/Q3(a)(iv)

3(a)(iv)	96% × (5000 / 42.0) × 72.0 = 8230 or 8229 (kg)	1
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32. O/N 22/P21/Q1(a)(ii)-(iii)

1(a)(ii)	M1 (weighted) average / mean mass of the isotopes / average mass of the atom(s) (of an element) M2 compared to (the mass of) the unified atomic mass unit	1
1(a)(iii)	69.723 = 68.926x + 70.925(1 – x) ∴ x = 0.6013 / 69.723 = $\frac{68.926x + 70.925(100 - x)}{100}$	1
	60.13%	1

33. O/N 22/P22/Q1(d)

1(d)(i)	phosphate(V)	1
1(d)(ii)	(molecular mass of NH ₄ MgPO ₄) = 137.3	1
	(245.3 – M _r (NH ₄ MgPO ₄) ÷ 18.0) = 6	1

34. O/N 22/P23/Q1(a)(ii)-(iii)

1(a)(ii)	M1 (weighted) average / mean mass of the isotopes / average mass of the atom(s) (of an element) M2 compared to (the mass of) the unified atomic mass unit	1
		1

1(a)(iii)	$69.723 = 68.926x + 70.925(1-x) \therefore x = 0.6013$ $/ 69.723 = \frac{68.926x + 70.925(100-x)}{100}$	1
	60.13%	1

35. M/J 22/P21/Q1(e)(i)-(ii)

1(e)(i)	1 / 12 (one twelfth) the mass of a carbon-12 / ^{12}C atom	1
1(e)(ii)	M1 correct expression relating A_r to the mass / % abundance of the three isotopes $24.31 = x \times 0.7899 + 24.99 \times 0.1000 + 25.98 \times 0.1101$ M2 correct answer to 4 sig figs atomic mass of X = 23.99	2

36. M/J 22/P21/Q4(a)

4(a)	M1 % / A_r for C H O M2 each % / A_r for C H O divided by the smallest value for % / A_r to give simplest whole number ratio / empirical formula M3 compare M_r from M2 ratio with 280 to deduce the actual molecular formula <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="text-align: center;">C</td> <td style="text-align: center;">H</td> <td style="text-align: center;">O</td> </tr> <tr> <td style="text-align: center;">$77.2 / 12 = 6.433$</td> <td style="text-align: center;">$11.4 / 1 = 11.4$</td> <td style="text-align: center;">$11.4 / 16 = 0.7125$</td> </tr> <tr> <td style="text-align: center;">9(.03)</td> <td style="text-align: center;">16</td> <td style="text-align: center;">1</td> </tr> </table> $M_r(\text{C}_9\text{H}_{16}\text{O}) = 140$ so molecular formula of V = $\text{C}_{18}\text{H}_{32}\text{O}_2$	C	H	O	$77.2 / 12 = 6.433$	$11.4 / 1 = 11.4$	$11.4 / 16 = 0.7125$	9(.03)	16	1	3
C	H	O									
$77.2 / 12 = 6.433$	$11.4 / 1 = 11.4$	$11.4 / 16 = 0.7125$									
9(.03)	16	1									

37. M/J 22/P21/Q4(b)(ii)

4(b)(ii)	$3.196 \text{ g Br}_2 = 3.196 / 159.8 (= 0.02 \text{ mol Br}_2)$ AND $2.8(00) / 280 (= 0.01 \text{ mol V})$ 2 alkene / 2 C=C (groups)	1
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38. M/J 22/P22/Q6(b)

6(b)	<table><tr><td>C</td><td>:</td><td>H</td><td>:</td><td>O</td></tr><tr><td>41.38 / 12</td><td></td><td>3.45 / 1</td><td></td><td>55.17 / 16</td></tr><tr><td>3.45</td><td></td><td>3.45</td><td></td><td>3.45</td></tr></table> (so C ₍₁₎ H ₍₁₎ O ₍₁₎)	C	:	H	:	O	41.38 / 12		3.45 / 1		55.17 / 16	3.45		3.45		3.45	1
C	:	H	:	O													
41.38 / 12		3.45 / 1		55.17 / 16													
3.45		3.45		3.45													

39. M/J 22/P23/Q1(f)(i)

1(f)(i)	$M_r = 63.5$ $(1.05 / 63.5) \times 6.022 \times 10^{23} = 9.958 \times 10^{21}$ OR 9.96×10^{21}	1
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